Sheet 2 Mon 12 Oct. 2026 *Determinant of two vectors; collinearity criterion*

1. Warm-up · Quick Calculate $2 \times (-3) - 4 \times 5$ and $(-1) \times 6 - 3 \times (-2)$.
2. Core · Method Calculate $\det(\vec{u}, \vec{v})$ for $\vec{u}\begin{pmatrix} 2 \\ -3 \end{pmatrix}$ and $\vec{v}\begin{pmatrix} -4 \\ 6 \end{pmatrix}$. What can be concluded?
3. Core + · True/false True or false? Justify. The vectors $\vec{u}\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\vec{v}\begin{pmatrix} 3 \\ 5 \end{pmatrix}$ are collinear.
4. Review · Construction Draw a circle with centre O and two perpendicular diameters $[AC]$ and $[BD]$. What type of quadrilateral is $ABCD$? Justify.
5. Thread · Quick Programme: choose a number; multiply it by 4; add 5. a. What is the result for 3? b. What is the expression in terms of the starting number x?

Sheet 3 Tue 13 Oct. 2026 *Collinearity and parallelism: the determinant method*

1. Warm-up · Quick Calculate the coordinates of $\overrightarrow{AB}$ for $A(1;-2)$ and $B(4;3)$.
2. Core · Method Are the points $A(1;2)$, $B(3;3)$ and $C(7;5)$ collinear?
3. Core + · Method For what value of k are the vectors $\vec{u}\begin{pmatrix} k \\ 3 \end{pmatrix}$ and $\vec{v}\begin{pmatrix} 4 \\ 6 \end{pmatrix}$ collinear?
4. Review · Quick Calculate 20 % of 350.
5. Thread · Translate A rectangle has length $x + 3$ and width x (in cm). Express its perimeter in terms of x and simplify.

Sheet 4 Thu 15 Oct. 2026 *Problem solving: vector, coordinate or geometric method*

1. Warm-up · Quick Complete: $ABCD$ is a parallelogram $\Leftrightarrow$ $\overrightarrow{AB} = \ldots$
2. Core · Method $A(0;0)$, $B(4;1)$, $C(5;4)$. Find D so that $ABCD$ is a parallelogram.
3. Core + · Write-up $A(0;0)$, $B(3;1)$, $C(-1;3)$. Prove that ABC is an isosceles right-angled triangle at A.
4. Review · Quick Write 450 000 in scientific notation.
5. Thread · Quick How many faces, vertices and edges does a cube have?

Lines in the plane

Sheet 5 Mon 2 Nov. 2026 *Direction vector of a line*

1. Warm-up · Quick Give a vector collinear to $\vec{u}\begin{pmatrix} 2 \\ -4 \end{pmatrix}$ with simpler integer coordinates.
2. Core · Method Give a direction vector of (AB) for $A(1;2)$ and $B(4;8)$.
3. Core + · MCQ A direction vector of the line through $A(0;1)$ and $B(2;-3)$ is:

   A. $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$ **B.** $\begin{pmatrix} 1 \\ -4 \end{pmatrix}$ **C.** $\begin{pmatrix} -4 \\ -2 \end{pmatrix}$ **D.** $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$
4. Review · Quick Marks: 8, 12, 12, 15, 18. Calculate the mean, the median and the range.
5. Thread · Property ABC is right-angled at A and I is the midpoint of $[BC]$. What can be said about IA, IB and IC? Why?

Sheet 6 Tue 3 Nov. 2026 *Cartesian equation of a line*

1. Warm-up · Quick Calculate $3x - 2y + 1$ for $x = 1$ and $y = -2$.
2. Core · Method The line d has equation $2x + 3y - 6 = 0$. Give a direction vector and a point of d.
3. Core + · True/false True or false? Justify. The lines $2x + 3y - 6 = 0$ and $4x + 6y + 1 = 0$ are parallel.
4. Review · Table Mode of transport of 30 students.

Transport	Bus	Bike	Walking	Car
Frequency	12	5	8	5

Give the proportion of students who come by bus and the angle of the "bike" sector in a pie chart.

5. Thread · Quick On a number line, A has coordinate -2 and B has coordinate 3. Give AB and the coordinate of the midpoint of $[AB]$.

Sheet 7 Thu 5 Nov. 2026 *Slope-intercept equation and gradient of a non-vertical line*

1. Warm-up · Quick Make y the subject of $2x + y - 5 = 0$.
2. Core · Method Give the gradient and the y-intercept of the line $3x - 2y + 4 = 0$.
3. Core + · MCQ The gradient of the line (AB) with $A(1;2)$ and $B(5;0)$ is:

   A. 2 **B.** $\frac{1}{2}$ **C.** -2 **D.** $-\frac{1}{2}$
4. Review · Quick Calculate $\frac{3}{4} + \frac{5}{6}$.
5. Thread · Problem Two gears have 12 and 18 teeth. After how many turns of each do they return together to their starting position?

Sheet 8 Mon 9 Nov. 2026 *Drawing a line given by an equation*

1. Warm-up · Quick For $y = -2x + 3$, calculate y when $x = 0$ and then when $x = 2$.
2. Core · Method Give two points that allow you to draw the line $2x + y - 4 = 0$.
3. Core + · Quick What is the line with equation $x = 3$? Give a direction vector. Does it have a gradient?
4. Review · Quick Compare $\frac{3}{7}$ and $\frac{4}{9}$.
5. Thread · Quick Solve in $\mathbb{R}$: $5x - 7 = 2x + 8$.

Sheet 9 Tue 10 Nov. 2026 *Finding the equation of a line: two points, a point and a direction vector, a point and a gradient*

1. Warm-up · Quick Gradient of the line through $(1;3)$ and $(3;7)$?
2. Core · Method Slope-intercept equation of the line with gradient 2 through $A(1;3)$.
3. Core + · Method Line through $A(1;2)$ with direction vector $\vec{u}\begin{pmatrix}2\\-1\end{pmatrix}$: Cartesian equation using the determinant of $\overrightarrow{AM}$ and $\vec{u}$.
4. Review · Quick Simplify $2^5 \times 2^{-3}$.
5. Thread · Table → formula An affine function f satisfies: $f(0) = -1$, $f(1) = 1$, $f(2) = 3$. Give $f(x)$.

Sheet 10 Thu 12 Nov. 2026 *Lines and affine functions: linking the representations*

1. Warm-up · Quick $f(x) = -3x + 4$. Calculate $f(2)$ and the preimage of 10.
2. Core · Method Find the affine function f such that $f(1) = 1$ and $f(3) = 5$.
3. Core + · True/false True or false? Justify. Every line in the plane is the graph of an affine function.
4. Review · Quick A fair 6-sided die is rolled. What is the probability of getting a multiple of 3?
5. Thread · Quick ABC is right-angled at A with $AB = 6$ cm and $AC = 8$ cm. Calculate BC.